

Lecture 2: Exploration and Exploitation

Rigorous Mathematical Notes

Based on lectures by Hado van Hasselt

1 Introduction and Recap

Reinforcement learning is the science of learning to make decisions. The fundamental components are:

- **Policy:** A mapping from states to actions (or distributions over actions)
- **Value function:** Expected cumulative reward from a state
- **Model:** Agent's representation of environment dynamics

The general problem involves taking into account time and consequences. Decisions affect the reward, the agent state, and environment state. Learning is *active*: decisions impact the data distribution.

Remark 1.1. In this lecture, we simplify the setting by assuming the environment has only a single state. This implies:

1. Actions no longer have long-term consequences in the environment
2. Actions still impact immediate reward
3. Other observations can be ignored

This simplified setting is known as the *multi-armed bandit* problem.

2 Exploration vs. Exploitation

Definition 2.1 (Exploration-Exploitation Trade-off). Learning agents must balance two competing objectives:

- **Exploitation:** Maximizing performance based on current knowledge
- **Exploration:** Gathering information to increase knowledge

Remark 2.2. The best long-term strategy may involve short-term sacrifices. We need to gather information to make the best overall decisions.

3 Formalizing the Problem

3.1 The Multi-Armed Bandit

Definition 3.1 (Multi-Armed Bandit). A *multi-armed bandit* is a tuple $(\mathcal{A}, \{\mathcal{R}_a\}_{a \in \mathcal{A}})$ where:

- $\mathcal{A}$ is a (known) finite set of actions (or “arms”)
- $\mathcal{R}_a$ is a probability distribution on rewards, given action a

Definition 3.2 (Interaction Protocol). At each time step $t = 1, 2, \dots$:

1. The agent selects an action $A_t \in \mathcal{A}$
2. The environment generates a reward $R_t \sim \mathcal{R}_{A_t}$

The goal is to maximize cumulative reward $\sum_{i=1}^t R_i$.

Definition 3.3 (Policy). A *policy* π is a distribution over $\mathcal{A}$. At time t , we denote by $\pi_t(a)$ the probability of selecting action a .

3.2 Values and Regret

Definition 3.4 (Action Value). The *action value* (or *Q-value*) for action a is the expected reward:

$$q(a) = \mathbb{E}[R_t \mid A_t = a] \quad (1)$$

Definition 3.5 (Optimal Value). The *optimal value* is defined as:

$$v_* = \max_{a \in \mathcal{A}} q(a) = \max_{a \in \mathcal{A}} \mathbb{E}[R_t \mid A_t = a] \quad (2)$$

Definition 3.6 (Action Regret). The *regret* (or *gap*) of an action a is:

$$\Delta_a = v_* - q(a) \quad (3)$$

Remark 3.7. The regret for any optimal action $a^* \in \arg \max_{a \in \mathcal{A}} q(a)$ is zero.

Definition 3.8 (Total Regret). The *total regret* up to time t is:

$$L_t = \sum_{n=1}^t (v_* - q(A_n)) = \sum_{n=1}^t \Delta_{A_n} \quad (4)$$

Proposition 3.9. *Maximizing cumulative reward is equivalent to minimizing total regret:*

$$\max \sum_{n=1}^t R_n \iff \min L_t \quad (5)$$

Definition 3.10 (Action Count). The *count* for action a up to time t is:

$$N_t(a) = \sum_{n=1}^t \mathbb{I}(A_n = a) \quad (6)$$

where $\mathbb{I}(\cdot)$ is the indicator function.

Proposition 3.11 (Regret Decomposition). *Total regret can be expressed in terms of action counts:*

$$L_t = \sum_{a \in \mathcal{A}} N_t(a) \Delta_a \quad (7)$$

Remark 3.12. Proposition 3.11 implies that a good algorithm ensures small counts $N_t(a)$ for actions with large regret Δ_a .

4 Algorithms

4.1 Action Value Estimation

Definition 4.1 (Sample Mean Estimator). The *sample mean estimator* for action value $q(a)$ is:

$$Q_t(a) = \frac{\sum_{n=1}^t \mathbb{I}(A_n = a) R_n}{\sum_{n=1}^t \mathbb{I}(A_n = a)} = \frac{\sum_{n=1}^t \mathbb{I}(A_n = a) R_n}{N_t(a)} \quad (8)$$

when $N_t(a) > 0$.

Proposition 4.2 (Incremental Update). The *sample mean estimator* can be computed incrementally:

$$Q_t(A_t) = Q_{t-1}(A_t) + \alpha_t \underbrace{(R_t - Q_{t-1}(A_t))}_{TD \text{ error}} \quad (9)$$

with $\alpha_t = \frac{1}{N_t(A_t)}$ and $N_t(A_t) = N_{t-1}(A_t) + 1$, where $N_0(a) = 0$ for all a .

For $a \neq A_t$: $Q_t(a) = Q_{t-1}(a)$.

Remark 4.3. Using a constant step size $\alpha_t = \alpha$ leads to *tracking* (exponential recency-weighted average) rather than averaging, which is useful in non-stationary environments.

4.2 Greedy Algorithm

Algorithm 4.4 (Greedy Policy). The *greedy policy* selects the action with highest estimated value:

$$A_t = \arg \max_{a \in \mathcal{A}} Q_t(a) \quad (10)$$

Equivalently, the policy is:

$$\pi_t(a) = \mathbb{I} \left(a = \arg \max_{b \in \mathcal{A}} Q_t(b) \right) \quad (11)$$

(assuming ties are broken deterministically).

Proposition 4.5 (Linear Regret of Greedy). The *greedy algorithm* can get stuck on a suboptimal action forever, resulting in linear expected total regret:

$$\mathbb{E}[L_t] = \Theta(t) \quad (12)$$

4.3 ε -Greedy Algorithm

Algorithm 4.6 (ε -Greedy Policy). For $\varepsilon \in (0, 1)$, the ε -greedy policy is:

$$\pi_t(a) = \begin{cases} 1 - \varepsilon + \frac{\varepsilon}{|\mathcal{A}|} & \text{if } Q_t(a) = \max_b Q_t(b) \\ \frac{\varepsilon}{|\mathcal{A}|} & \text{otherwise} \end{cases} \quad (13)$$

Equivalently:

- With probability $1 - \varepsilon$: select greedy action $\arg \max_a Q_t(a)$
- With probability ε : select a uniformly random action

Proposition 4.7 (Linear Regret of ε -Greedy). ε -greedy with constant $\varepsilon > 0$ has linear expected total regret:

$$\mathbb{E}[L_t] = \Theta(t) \quad (14)$$

because it continues to explore uniformly forever.

4.4 Policy Gradient Methods

Definition 4.8 (Softmax Policy). Given *action preferences* $H_t(a) \in \mathbb{R}$ for each $a \in \mathcal{A}$, the *softmax policy* is:

$$\pi_\theta(a) = \frac{e^{H_t(a)}}{\sum_{b \in \mathcal{A}} e^{H_t(b)}} \quad (15)$$

where $\theta = (H_t(a))_{a \in \mathcal{A}}$ are the policy parameters.

Remark 4.9. The preferences $H_t(a)$ are not value estimates; they are learnable policy parameters.

Definition 4.10 (Policy Gradient Objective). The goal is to maximize the expected reward under the policy:

$$J(\theta) = \mathbb{E}[R_t \mid \pi_\theta] = \sum_{a \in \mathcal{A}} \pi_\theta(a) q(a) \quad (16)$$

Theorem 4.11 (Policy Gradient Theorem for Bandits). *The gradient of the expected reward is:*

$$\nabla_\theta \mathbb{E}[R_t \mid \pi_\theta] = \mathbb{E}[R_t \nabla_\theta \log \pi_\theta(A_t)] \quad (17)$$

Proof. Using the log-likelihood trick (REINFORCE, Williams 1992):

$$\nabla_\theta \mathbb{E}[R_t \mid \pi_\theta] = \nabla_\theta \sum_a \pi_\theta(a) \underbrace{\mathbb{E}[R_t \mid A_t = a]}_{=q(a)} \quad (18)$$

$$= \sum_a q(a) \nabla_\theta \pi_\theta(a) \quad (19)$$

$$= \sum_a q(a) \frac{\pi_\theta(a)}{\pi_\theta(a)} \nabla_\theta \pi_\theta(a) \quad (20)$$

$$= \sum_a \pi_\theta(a) q(a) \frac{\nabla_\theta \pi_\theta(a)}{\pi_\theta(a)} \quad (21)$$

$$= \mathbb{E} \left[q(A_t) \frac{\nabla_\theta \pi_\theta(A_t)}{\pi_\theta(A_t)} \right] \quad (22)$$

$$= \mathbb{E}[R_t \nabla_\theta \log \pi_\theta(A_t)] \quad (23)$$

where the last equality uses $\mathbb{E}[R_t \mid A_t] = q(A_t)$. $\square$

Algorithm 4.12 (REINFORCE for Bandits). Update policy parameters via stochastic gradient ascent:

$$\theta \leftarrow \theta + \alpha R_t \nabla_\theta \log \pi_\theta(A_t) \quad (24)$$

Proposition 4.13 (Gradient for Softmax Policy). *For the softmax policy, the gradient with respect to preference $H_t(a)$ is:*

$$\frac{\partial \log \pi_t(A_t)}{\partial H_t(a)} = \mathbb{I}(a = A_t) - \pi_t(a) \quad (25)$$

Corollary 4.14 (Softmax Policy Update). *The update rules for the softmax policy are:*

$$H_{t+1}(A_t) = H_t(A_t) + \alpha R_t (1 - \pi_t(A_t)) \quad (26)$$

$$H_{t+1}(a) = H_t(a) - \alpha R_t \pi_t(a) \quad \text{for } a \neq A_t \quad (27)$$

Remark 4.15. Preferences for actions with higher rewards increase more (or decrease less), making them more likely to be selected again.

5 Theoretical Bounds

5.1 Lower Bound on Regret

Theorem 5.1 (Lai-Robbins Lower Bound). *For any algorithm and any multi-armed bandit problem, the asymptotic expected total regret satisfies:*

$$\liminf_{t \rightarrow \infty} \frac{\mathbb{E}[L_t]}{\log t} \geq \sum_{a: \Delta_a > 0} \frac{\Delta_a}{\text{KL}(\mathcal{R}_a \| \mathcal{R}_{a^*})} \quad (28)$$

where $\text{KL}(\mathcal{R}_a \| \mathcal{R}_{a^*})$ is the Kullback-Leibler divergence between the reward distributions of action a and an optimal action a^* .

Remark 5.2. For distributions where $\text{KL}(\mathcal{R}_a \| \mathcal{R}_{a^*}) \propto \Delta_a^2$ (e.g., Gaussian with fixed variance), the lower bound becomes:

$$\liminf_{t \rightarrow \infty} \frac{\mathbb{E}[L_t]}{\log t} \geq \sum_{a: \Delta_a > 0} \frac{1}{\Delta_a} \quad (29)$$

Corollary 5.3. *The expected total regret must grow at least logarithmically in t . No algorithm can achieve sublogarithmic regret uniformly across all bandit instances.*

5.2 Concentration Inequalities

Theorem 5.4 (Hoeffding's Inequality). *Let $X_1, \dots, X_n$ be i.i.d. random variables with $X_i \in [0, 1]$ and true mean $\mu = \mathbb{E}[X]$. Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ be the sample mean. Then for all $u > 0$:*

$$\Pr(\bar{X}_n + u \leq \mu) \leq e^{-2nu^2} \quad (30)$$

and by symmetry:

$$\Pr(\bar{X}_n - u \geq \mu) \leq e^{-2nu^2} \quad (31)$$

Corollary 5.5 (Confidence Bound for Bandits). *For bandits with bounded rewards $R_t \in [0, 1]$:*

$$\Pr(Q_t(a) + U_t(a) \leq q(a)) \leq e^{-2N_t(a)U_t(a)^2} \quad (32)$$

where $Q_t(a)$ is the sample mean and $U_t(a)$ is the confidence width.

6 Optimism in the Face of Uncertainty

Definition 6.1 (Principle of Optimism). *The principle of optimism in the face of uncertainty states: when uncertain about action values, act as if the most optimistic scenario is true. This naturally balances exploration and exploitation because:*

- Actions with high estimated values are selected (exploitation)
- Actions with high uncertainty are selected (exploration)

6.1 Upper Confidence Bound (UCB)

Definition 6.2 (Upper Confidence Bound). An *upper confidence bound* $U_t(a)$ for action a at time t satisfies:

$$q(a) \leq Q_t(a) + U_t(a) \quad \text{with high probability} \quad (33)$$

Algorithm 6.3 (UCB Action Selection). Select the action maximizing the upper confidence bound:

$$A_t = \arg \max_{a \in \mathcal{A}} (Q_t(a) + U_t(a)) \quad (34)$$

Remark 6.4. The uncertainty $U_t(a)$ should depend on the count $N_t(a)$:

- Small $N_t(a) \Rightarrow$ large $U_t(a)$ (uncertain estimate)
- Large $N_t(a) \Rightarrow$ small $U_t(a)$ (accurate estimate)

Proposition 6.5 (Deriving the UCB). *Setting the failure probability to p and solving:*

$$e^{-2N_t(a)U_t(a)^2} = p \implies U_t(a) = \sqrt{\frac{-\log p}{2N_t(a)}} \quad (35)$$

Choosing $p = 1/t$ (probability decreasing with time):

$$U_t(a) = \sqrt{\frac{\log t}{2N_t(a)}} \quad (36)$$

Algorithm 6.6 (UCB1). The *UCB1* algorithm selects:

$$A_t = \arg \max_{a \in \mathcal{A}} \left(Q_t(a) + c \sqrt{\frac{\log t}{N_t(a)}} \right) \quad (37)$$

where $c > 0$ is a hyperparameter.

Theorem 6.7 (UCB Regret Bound (Auer et al., 2002)). *UCB1 with $c = \sqrt{2}$ achieves logarithmic expected total regret:*

$$\mathbb{E}[L_t] \leq 8 \sum_{a: \Delta_a > 0} \frac{\log t}{\Delta_a} + O \left(\sum_a \Delta_a \right) \quad \forall t \quad (38)$$

Corollary 6.8. *UCB achieves order-optimal regret matching the Lai-Robbins lower bound up to constant factors.*

Lemma 6.9 (Key Insight for UCB Analysis). *For UCB, either Δ_a is small or $N_t(a)$ is small. Specifically:*

$$\Delta_a N_t(a) \leq O(\log t) \quad \forall a \quad (39)$$

7 Bayesian Approaches

7.1 Bayesian Bandits

Definition 7.1 (Bayesian Bandit). In the *Bayesian* approach, we maintain a posterior distribution over action values:

$$p(q(a) \mid \theta_t) \quad (40)$$

where θ_t summarizes the history of observations. The distribution $p(q(a) = x \mid \theta_t)$ represents our *belief* that $q(a) = x$.

Example 7.2 (Bernoulli Bandit with Beta Prior). Consider a Bernoulli bandit where $R_t \in \{0, 1\}$ with $\Pr(R_t = 1 \mid A_t = a) = \mu_a$.

Prior: Uniform distribution on $[0, 1]$, i.e., $\text{Beta}(1, 1)$.

Posterior: After observing rewards, the posterior is $\text{Beta}(\alpha_a, \beta_a)$ where:

- $\alpha_a = 1 + (\text{number of } R = 0 \text{ outcomes for action } a)$
- $\beta_a = 1 + (\text{number of } R = 1 \text{ outcomes for action } a)$

Update:

$$\alpha_{A_t} \leftarrow \alpha_{A_t} + 1 \quad \text{when } R_t = 0 \quad (41)$$

$$\beta_{A_t} \leftarrow \beta_{A_t} + 1 \quad \text{when } R_t = 1 \quad (42)$$

Algorithm 7.3 (Bayesian UCB). Estimate upper confidence bounds from the posterior:

$$U_t(a) = c \cdot \sigma_t(a) \quad (43)$$

where $\sigma_t(a)$ is the posterior standard deviation. Select:

$$A_t = \arg \max_{a \in \mathcal{A}} (Q_t(a) + c \cdot \sigma_t(a)) \quad (44)$$

where $Q_t(a)$ is the posterior mean.

7.2 Thompson Sampling

Definition 7.4 (Probability Matching). *Probability matching* selects action a according to the posterior probability that a is optimal:

$$\pi_t(a) = \Pr \left(q(a) = \max_{a'} q(a') \mid \mathcal{H}_{t-1} \right) \quad (45)$$

where $\mathcal{H}_{t-1}$ denotes the history up to time $t - 1$.

Remark 7.5. Probability matching is optimistic in the face of uncertainty: actions have higher selection probability when either the estimated value is high or the uncertainty is high.

Algorithm 7.6 (Thompson Sampling). Thompson Sampling (Thompson, 1933) implements probability matching via sampling:

1. Sample $\tilde{Q}_t(a) \sim p_t(q(a))$ for all $a \in \mathcal{A}$
2. Select $A_t = \arg \max_{a \in \mathcal{A}} \tilde{Q}_t(a)$

Proposition 7.7 (Thompson Sampling as Probability Matching). *Thompson Sampling implements probability matching:*

$$\pi_t(a) = \mathbb{E} \left[\mathbb{I} \left(\tilde{Q}_t(a) = \max_{a'} \tilde{Q}_t(a') \right) \right] \quad (46)$$

$$= \Pr \left(q(a) = \max_{a'} q(a') \right) \quad (47)$$

Theorem 7.8 (Optimality of Thompson Sampling). *For Bernoulli bandits, Thompson Sampling achieves the Lai-Robbins lower bound on regret, and is therefore asymptotically optimal.*

8 Planning to Explore: Information State Space

8.1 Bandits as Sequential Decision Problems

Definition 8.1 (Information State). The *information state* S_t summarizes the agent's history and knowledge at time t . Each action A_t causes a transition to a new information state S_{t+1} with probability $p(S_{t+1} | A_t, S_t)$.

Remark 8.2. Even in bandits (single environment state), actions affect the future through learning. The information state evolves as the agent gains knowledge.

Proposition 8.3. The bandit problem can be reformulated as a Markov Decision Process (MDP) over information states, where:

- *State space:* all possible information states
- *Actions:* bandit arms
- *Transitions:* information state updates based on observed rewards
- *Rewards:* same as the bandit rewards

Example 8.4 (Bernoulli Bandit Information State). For a Bernoulli bandit with $|\mathcal{A}|$ arms:

- Information state: $I = (\alpha, \beta)$ where α_a and β_a are the Beta distribution parameters for each arm a
- α_a counts pulls of arm a with reward 0
- β_a counts pulls of arm a with reward 1

8.2 Bayes-Adaptive Reinforcement Learning

Definition 8.5 (Bayes-Adaptive MDP). The *Bayes-adaptive MDP* formulation solves the information state MDP optimally, yielding policies that optimally trade off exploration and exploitation with respect to the prior distribution.

Remark 8.6. Bayes-adaptive RL can be extended to full MDPs by also maintaining beliefs over transition dynamics. However, the information state space is typically infinite, making exact solutions intractable. Approximate methods are required for practical applications.

9 Summary

Algorithm	Expected Regret	Key Property
Greedy	$\Theta(t)$ (linear)	No exploration
ε -Greedy (constant ε)	$\Theta(t)$ (linear)	Uniform exploration
UCB1	$O(\log t)$	Optimism via confidence bounds
Thompson Sampling	$O(\log t)$	Posterior sampling

Theorem 9.1 (Summary of Regret Bounds). • **Lower bound (Lai-Robbins):** $\mathbb{E}[L_t] = \Omega(\log t)$ for any algorithm

- **Upper bound (UCB, Thompson):** $\mathbb{E}[L_t] = O(\log t)$ is achievable
- **Conclusion:** Logarithmic regret is optimal and achievable